

Significant Figures with Uncertainties: Rules, Cases and Verdict

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Introduction

Reporting measured or computed values with uncertainty requires matching the precision of the value to the precision of the uncertainty. This document provides the core rules, correct and incorrect examples, and a checklist for use before meetings or presentations. The key principle is to judge each $\text{value} \pm \text{uncertainty}$ pair independently, never by comparing decimal places across rows or columns.

1 Core Rules

Rule 1: Match decimal places. The uncertainty determines the decimal place of the value. Round the value so its last decimal place matches the uncertainty's last decimal place.

Rule 2: Uncertainty usually has 1 sig fig. Uncertainties are reported to one significant figure (e.g., 0.07, 0.1, 5, 30). Exception: if the uncertainty starts with 1, you may use 2 sig figs (e.g., 12, 15).

Rule 3: Judge each pair independently. Each $\text{value} \pm \text{uncertainty}$ pair stands on its own. Do not compare decimal places across rows or columns. A different value in the same dataset can have a different number of decimal places if its uncertainty justifies it.

Why it works: Uncertainty tells the reader which digit is uncertain. If uncertainty is 0.07, digits below the hundredths place are meaningless noise. The value's precision must match what the uncertainty says.

2 Right Cases

The following $\text{value} \pm \text{uncertainty}$ pairs follow the rules correctly:

- 3.94 ± 0.05 : Uncertainty is at hundredths; value stops at hundredths. Both have 2 decimal places.
- 5.11 ± 0.07 : Uncertainty is at hundredths; value stops at hundredths. Both have 2 decimal places.
- 5.6 ± 0.1 : Uncertainty is at tenths; value stops at tenths. Both have 1 decimal place.
- 42 ± 3 : Uncertainty is at ones; value is a whole number. No decimal places needed.
- 2.47 ± 0.08 : Uncertainty starts at hundredths; value stops at hundredths. Matched.
- 1.24 ± 0.15 : Uncertainty starts with 1, so 2 sig figs OK; value has 2 decimals. Matched.

3 Wrong Cases and Fixes

Mismatch: Value has more decimal places than uncertainty

- **Wrong:** 3.94 ± 0.3 **Problem:** Uncertainty is at tenths, but value goes to hundredths. **Fixed:** 3.9 ± 0.3
- **Wrong:** 5.11 ± 0.1 **Problem:** Uncertainty is at tenths, but value goes to hundredths. **Fixed:** 5.1 ± 0.1
- **Wrong:** 42.6 ± 2 **Problem:** Uncertainty is at ones, but value goes to tenths. **Fixed:** 43 ± 2 (or 42 ± 2)

Mismatch: Value has fewer decimal places than uncertainty

- **Wrong:** 5.6 ± 0.07 **Problem:** Uncertainty is at hundredths, but value stops at tenths. **Fixed:** 5.60 ± 0.07

Not a mistake: Different decimal places in different rows

- Row A: 3.4 ± 0.5 (1 decimal place)
- Row B: 3.94 ± 0.05 (2 decimal places)

This is **correct**. Each pair follows its own rule based on its uncertainty. Do not force all values in a column to have the same decimal places.

4 Gray Areas and Judgment Calls

When to keep high precision in uncertainty

If you are tempted to round an uncertainty to fewer sig figs, check the percent change:

- Uncertainty 0.07 rounded to 0.1: change $\approx 43\%$ (huge). Keep 0.07.
- Uncertainty 0.08 rounded to 0.1: change $\approx 25\%$ (too much). Keep 0.08.
- Uncertainty 0.15 rounded to 0.2: change $\approx 33\%$ (too much). Keep 0.15.
- Uncertainty 0.04 rounded to 0.05: change $\approx 25\%$ (too much). Keep 0.04.

Comparing datasets with different uncertainties

One method yields a value of $2.43 \pm 0.08 \text{ cm}^2/\text{s}$. Another method gives $2.4 \pm 0.1 \text{ cm}^2/\text{s}$. The decimal places differ (2 vs 1) because the uncertainties differ. **This is not a mistake**. Each value reports what its own uncertainty demands. The first measurement is more precise; the second is less precise. This is the correct way to represent them.

5 Final Verdict Checklist

Before any meeting or presentation, scan your table and verify:

1. **For each row:** Does the value's last decimal place match the uncertainty's last decimal place?
2. **For each row:** Is the uncertainty 1 sig fig (or 2 if it starts with 1)?
3. **Across rows:** Are different decimal places OK? Answer: yes, if each matches its own uncertainty.
4. **Spot check:** Pick one value $\pm$ uncertainty pair and ask: "Would halving the digits below the uncertainty's decimal place change the meaning?" If no, you're good.

Common mistake to avoid: Do not enforce a rule like "all values in this column must have the same number of decimal places." That rule is wrong. Each pair stands alone.

Summary

The golden rule is simple: **the uncertainty determines the decimal place of the value**. Once you internalize that, the rest follows. Judge each pair independently. When in doubt, compare to the uncertainty and match its decimal place. This ensures your data presentation is both scientifically sound and immediately clear to your audience.